

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 5: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The quadratic formula

1. Solve $2x^2 - 5x - 4 = 0$, giving the answers correct to three significant figures. **ICSE 2025, 4 marks**

SOLUTION

$$a = 2, b = -5, c = -4. D = 25 + 32 = 57 \text{ and } \sqrt{57} \approx 7.550.$$

$$x = \frac{5 \pm 7.550}{4}, \text{ so } x = \frac{12.550}{4} = 3.1375 \text{ or } x = \frac{-2.550}{4} = -0.6375$$

Answer: $x = 3.14$ or $x = -0.637$ (using $\sqrt{57} = 7.5498$, the second root is -0.63746)

2. Solve $2x^2 - 10x + 5 = 0$, giving the answers correct to three significant figures. **ICSE 2024, 3 marks**

SOLUTION

$$a = 2, b = -10, c = 5. D = 100 - 40 = 60 \text{ and } \sqrt{60} \approx 7.746.$$

$$x = \frac{10 \pm 7.746}{4}, \text{ so } x = \frac{17.746}{4} = 4.4365 \text{ or } x = \frac{2.254}{4} = 0.5635$$

Answer: $x = 4.44$ or $x = 0.564$

3. Solve $x^2 + 4x - 8 = 0$, giving the answers correct to one decimal place. **ICSE 2023, 4 marks**

SOLUTION

$$a = 1, b = 4, c = -8. D = 16 + 32 = 48 \text{ and } \sqrt{48} \approx 6.928.$$

$$x = \frac{-4 \pm 6.928}{2}, \text{ so } x = 1.464 \text{ or } x = -5.464$$

Answer: $x = 1.5$ or $x = -5.5$

4. Solve $x^2 - 7x + 3 = 0$, giving the answers correct to two decimal places. **ICSE 2020, 3 marks**

SOLUTION

$$a = 1, b = -7, c = 3. D = 49 - 12 = 37 \text{ and } \sqrt{37} \approx 6.083.$$

$$x = \frac{7 \pm 6.083}{2}, \text{ so } x = 6.5415 \text{ or } x = 0.4585$$

Answer: $x = 6.54$ or $x = 0.46$

5. Solve $x^2 - 4x - 8 = 0$, giving the answers correct to three significant figures. **ICSE 2019, 3 marks**

SOLUTION

$$D = 16 + 32 = 48 \text{ and } \sqrt{48} \approx 6.928$$

$$x = \frac{4 \pm 6.928}{2}, \text{ so } x = 5.464 \text{ or } x = -1.464$$

Answer: $x = 5.46$ or $x = -1.46$

6. Solve $x^2 + 7x = 7$, giving the answers correct to two decimal places. **ICSE 2018, 4 marks**

SOLUTION

$$\text{Write it as } x^2 + 7x - 7 = 0. D = 49 + 28 = 77 \text{ and } \sqrt{77} \approx 8.775.$$

$$x = \frac{-7 \pm 8.775}{2}, \text{ so } x = 0.8875 \text{ or } x = -7.8875$$

Answer: $x = 0.89$ or $x = -7.89$

7. Solve $4x^2 - 5x - 3 = 0$, giving the answers correct to two decimal places. **ICSE 2017, 4 marks**

SOLUTION

$$D = 25 + 48 = 73 \text{ and } \sqrt{73} \approx 8.544$$

$$x = \frac{5 \pm 8.544}{8}, \text{ so } x = 1.693 \text{ or } x = -0.443$$

Answer: $x = 1.69$ or $x = -0.44$

2 Nature of the roots

8. The equation $3x^2 + \sqrt{7}x + 2 = 0$ has: (a) two equal real roots (b) two distinct real roots (c) more than two real roots (d) no real roots **ICSE 2025, 1 mark**

SOLUTION

$$D = b^2 - 4ac = 7 - 24 = -17, \text{ which is negative.}$$

Answer: (d)

9. The roots of $px^2 - qx + r = 0$ are real and equal when: (a) $p^2 = 4qr$ (b) $q^2 = 4pr$ (c) $-q^2 = 4pr$ (d) $p^2 > 4qr$ **ICSE 2024, 1 mark**

SOLUTION

$$\text{Equal roots need } D = 0: (-q)^2 - 4pr = 0, \text{ so } q^2 = 4pr.$$

Answer: (b)

10. Find the values of k for which $x^2 + 4kx + (k^2 - k + 2) = 0$ has equal roots. **ICSE 2018, 3 marks**

SOLUTION

Equal roots need $D = 0$: $(4k)^2 - 4(k^2 - k + 2) = 0$

$12k^2 + 4k - 8 = 0$, that is $3k^2 + k - 2 = 0$

$(3k - 2)(k + 1) = 0$

Answer: $k = \frac{2}{3}$ or $k = -1$

3 Standard form and roots

11. 3 is a root of $x^2 - px + 3 = 0$. Find p . (a) 4 (b) 3 (c) 5 (d) 2 **ICSE 2023, 1 mark**

SOLUTION

Put $x = 3$: $9 - 3p + 3 = 0$, so $p = 4$.

Answer: (a)

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.